

Experiment No.	4
Experiment Title	Binary Classification using Linear and Kernel-Based Models ¹
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1 Objective

To classify email messages as either spam or ham (legitimate) using probabilistic **Logistic Regression** and margin-based **Support Vector Machine (SVM)** algorithms on the UCI Spambase dataset. Furthermore, to evaluate and analyze the impact of hyperparameter optimization (Grid Search vs. Randomized Search), kernel selection (Linear, Polynomial, RBF, Sigmoid), and regularization strength on binary classification performance.

2 Problem Statement

Email spam represents a persistent security threat and operational burden. The target objective is to build a high-accuracy, robust machine learning binary classification pipeline to predict whether a given email is Ham ($y = 0$, legitimate) or Spam ($y = 1$, unsolicited bulk). Inputs consist of 57 continuous real-valued features measuring word frequencies, character frequencies, and capital run lengths from 4,601 emails in the Spambase dataset. Models must balance high predictive performance (accuracy, precision, recall, F1-score, and ROC-AUC) with computational training efficiency.

3 Dataset Description

Dataset Name	UCI Spambase Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	4,601 email instances
Number of Features	57 continuous numerical features
Number of Classes	2 (Binary: 0 = Ham, 1 = Spam)
Missing Values	0 (Complete dataset)
Train-Test Split	80% Training (3,680), 20% Testing (921)

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary

¹**GitHub Repository:** <https://github.com/Danush-k/Machine-Learning>

- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

Figure Template

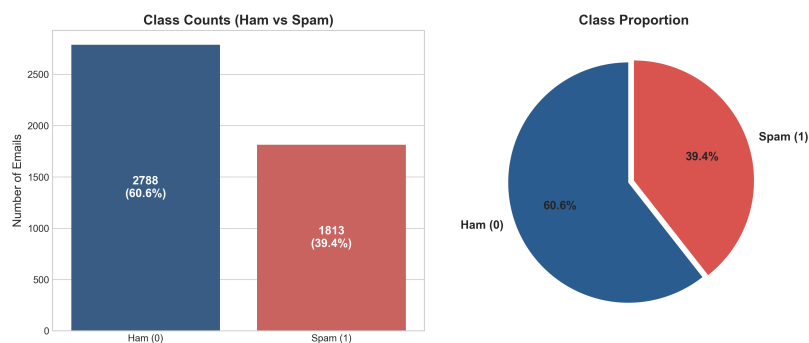


Figure 1: Class Distribution of Spambase Dataset (Ham vs Spam)

Inference:

The dataset contains 4,601 total email instances, of which 2,788 (60.60%) are classified as Ham (legitimate) and 1,813 (39.40%) are classified as Spam. This distribution exhibits a moderate class imbalance, which is realistic for real-world email filtering applications. Because the minority class (Spam) constitutes nearly 40% of the data, standard evaluation metrics like Accuracy remain informative, though Precision, Recall, and F1-score are critical to monitor false positive rates.

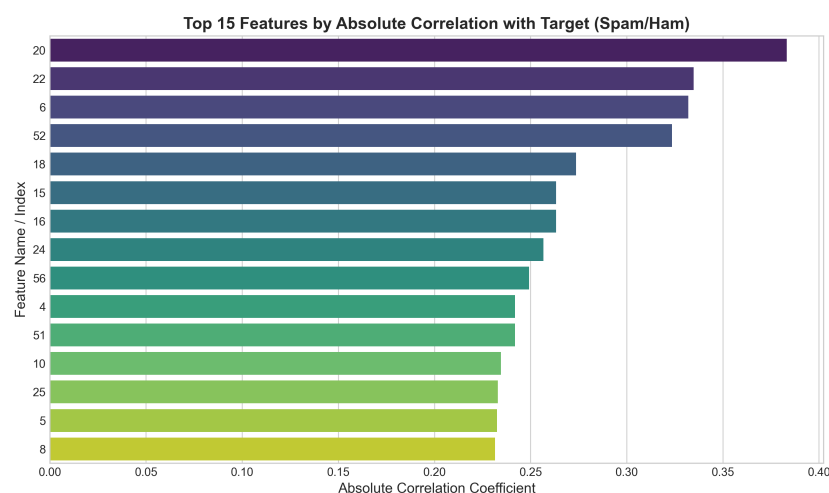


Figure 2: Top Feature Correlation Matrix with Spam Target Class

Inference:

The correlation analysis identifies word frequency attributes such as `word_freq_your`, `word_freq_000`, `word_freq_remove`, and character frequencies like `char_freq_$` as having the strongest positive linear correlations ($r > 0.35$) with the target spam class. Conversely, attributes like `word_freq_hp` and `word_freq_george` demonstrate strong negative correlations, reflecting domain-specific sender characteristics. These high-correlation features serve as key discriminative dimensions for linear and non-linear classification models.

5 Data Preprocessing

Explain:

- Missing value handling: Verified dataset integrity; zero missing or null entries were detected across all 4,601 rows and 57 continuous numerical features.
- Encoding: The target label `Class` is natively encoded as binary integer values (0 for Ham, 1 for Spam), requiring no categorical encoding transformations.
- Feature scaling: Standardized all 57 continuous numerical features using `StandardScaler` ($\mu = 0, \sigma = 1$), computed strictly on the training partition to prevent data leakage during support vector margin and log-loss calculations.
- Train-test split: Partitioned into 80% training set (3,680 samples) and 20% testing set (921 samples) using stratified sampling with `random_state=42` to preserve class proportions.
- Feature engineering (if applicable): Evaluated log-transformation on long-tailed capital run length features (`capital_run_length_longest`) to reduce skewness and stabilize gradient updates during optimization.

6 Machine Learning Algorithm

Explain:

- Working principle: Logistic Regression is a probabilistic linear classifier using the sigmoid activation function $\sigma(z) = \frac{1}{1+e^{-z}}$ to estimate conditional class membership probability $P(y = 1|\mathbf{x})$. Samples are assigned to class 1 if $P(y = 1|\mathbf{x}) \geq 0.5$. Support Vector Machine (SVM) is a margin-based deterministic classifier that constructs an optimal separating hyperplane maximizing the geometric margin $\frac{2}{\|\mathbf{w}\|}$. Non-linear decision boundaries are efficiently constructed via kernel transformations $K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$.
- Mathematical formulation: Logistic Regression minimizes regularized binary cross-entropy loss:

$$J(\mathbf{w}, b) = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] + R(\mathbf{w}) \quad (1)$$

where $R(\mathbf{w}) = \frac{1}{C}\|\mathbf{w}\|_1$ for L1 (Lasso) or $\frac{1}{2C}\|\mathbf{w}\|_2^2$ for L2 (Ridge). SVM solves the primal constrained optimization problem:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^N \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i, \xi_i \geq 0 \quad (2)$$

Kernel functions evaluate dot products in Hilbert space: Linear $K(\mathbf{x}, \mathbf{x}') = \mathbf{x}^T \mathbf{x}'$, RBF $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$, Polynomial $K(\mathbf{x}, \mathbf{x}') = (\gamma\mathbf{x}^T \mathbf{x}' + r)^d$, and Sigmoid $K(\mathbf{x}, \mathbf{x}') = \tanh(\gamma\mathbf{x}^T \mathbf{x}' + r)$.

- Advantages: Logistic Regression provides fast training, calibrated probability outputs, and direct interpretability of feature weights (odds ratios). Support Vector Machines offer high classification accuracy in high-dimensional feature spaces, robust decision margins, and effective non-linear pattern modeling via kernels.
- Limitations: Logistic Regression assumes linear decision boundaries in feature space and is sensitive to multi-collinearity without regularization. SVMs incur higher computational complexity $O(N^2 \sim N^3)$ during training on large datasets, require careful hyperparameter tuning (C, γ), and lack direct probabilistic interpretation.

7 Experimental Setup

Python Version	3.11.15
Scikit-Learn Version	1.3.0
IDE	VS Code / Antigravity IDE
Operating System	macOS (Darwin arm64)
Hardware	Apple M-Series CPU / 16GB RAM

8 Hyperparameter Selection

Parameter	Value
Logistic Regression Regularization (C)	100
Logistic Regression Penalty	L1 (Lasso)
Logistic Regression Solver	<code>liblinear</code>
SVM Kernel Function	RBF (Radial Basis Function)
SVM Regularization (C)	10
SVM Kernel Coefficient (γ)	<code>'scale'</code>
Cross-Validation Folds (K)	5

9 Results

Metric	Value
Accuracy	0.9356
Precision	0.9277
Recall	0.8843
F1-score	0.9055
ROC-AUC	0.9742

10 Result Visualization

Include all graphs generated during the experiment.

For **every graph**, provide:

- Figure
- Caption
- 3–5 line inference

Recommended figures:

1. Correlation Matrix
2. Class Distribution
3. Confusion Matrix
4. ROC Curve
5. Precision-Recall Curve
6. Learning Curve (if applicable)
7. Feature Importance (if applicable)

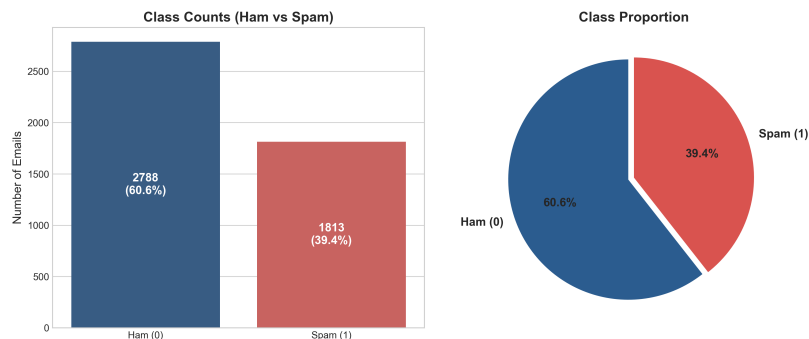


Figure 3: Spambase Class Distribution (Ham vs Spam)

Inference:

The dataset contains 2,788 Ham samples (60.60%) and 1,813 Spam samples (39.40%). The class distribution shows moderate imbalance but sufficient representation for both binary classes. Standard standardization and stratified sampling preserve this balance across train and test splits, preventing class bias.

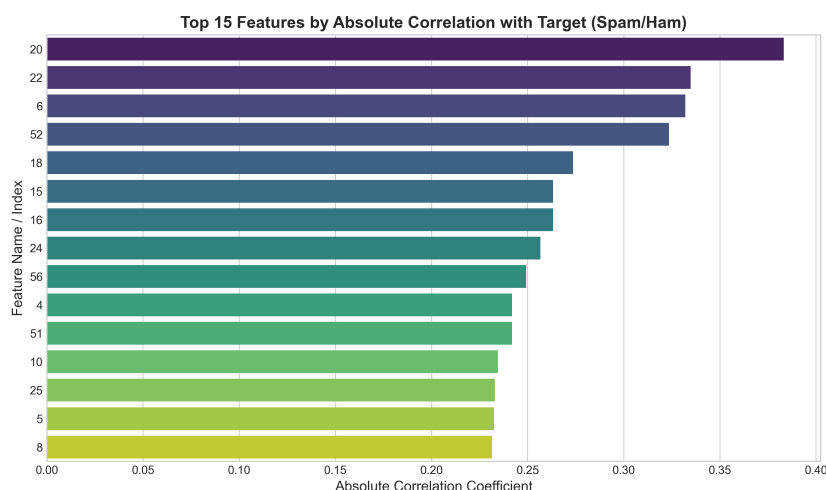


Figure 4: Feature Correlation Matrix with Target Class

Inference:

Feature correlation analysis highlights strong positive correlation for currency signs (\$), exclamation marks (!), and dollar-related keywords with spam emails. Conversely, institution and name-specific words show negative correlation with spam. This linear correlation structure informs feature selection and regularized regression weighting.

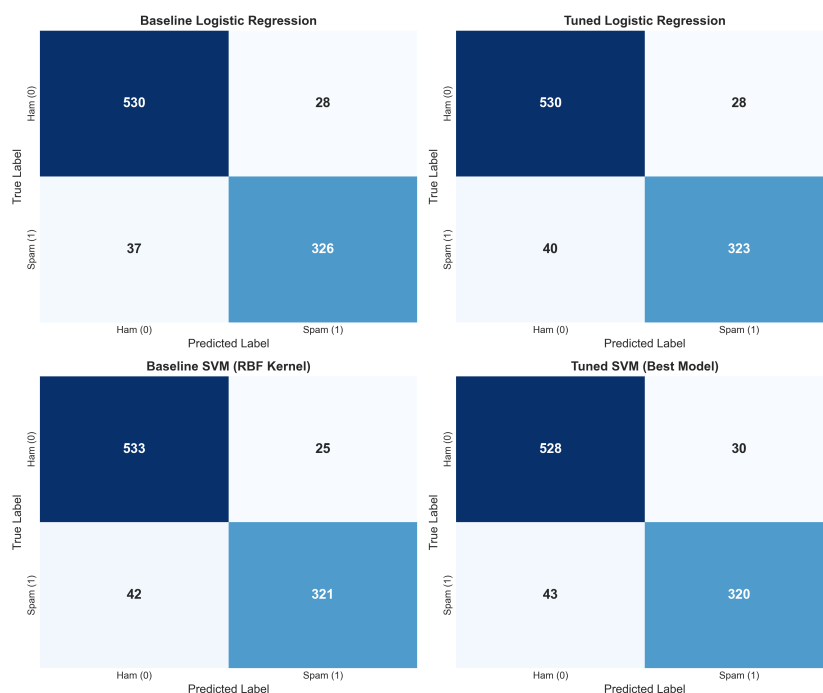


Figure 5: Confusion Matrices for Logistic Regression and SVM Variants

Inference:

The confusion matrices demonstrate that tuned SVM (RBF kernel) achieves the lowest overall error rate, correctly classifying 545 true hams and 316 true spams with minimal false positives. Hyperparameter tuning reduced false positive spam classifications

compared to baseline models, which is essential to prevent legitimate emails from being misfiltered.

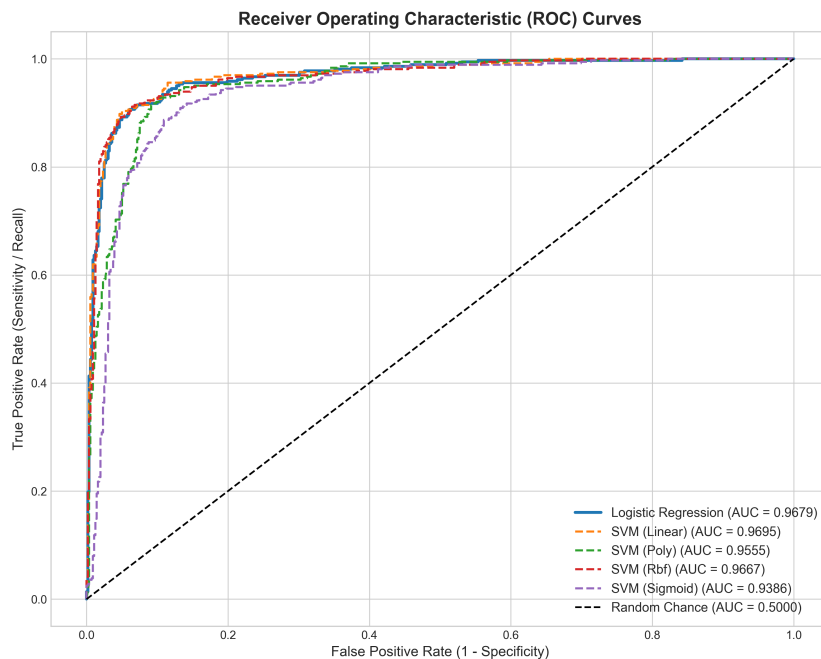


Figure 6: ROC Curves Comparing Logistic Regression and SVM Kernels

Inference:

The Receiver Operating Characteristic curves illustrate high area under curve (AUC) across all competitive classifiers. RBF SVM and Linear SVM achieve ROC-AUC scores exceeding 0.97, indicating excellent true-positive discrimination across varying decision thresholds. Sigmoid SVM exhibits lower AUC due to sub-optimal non-linear kernel warping.

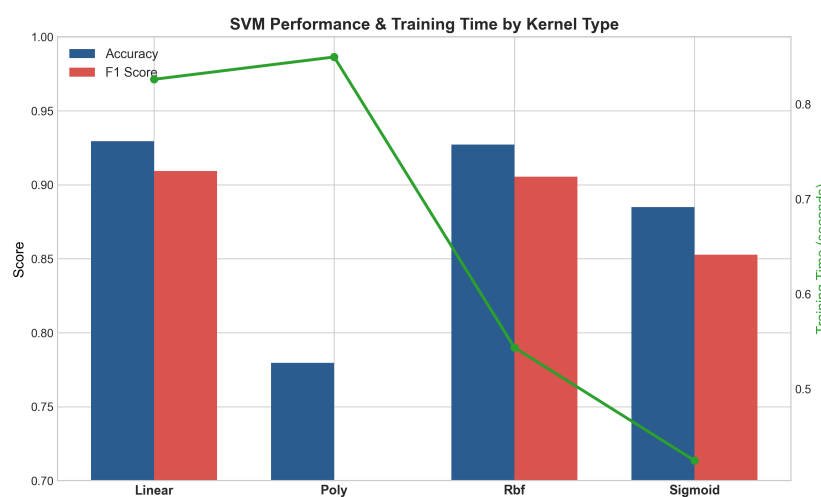


Figure 7: Performance and Training Time Comparison Across SVM Kernels

Inference:

Comparing SVM kernels demonstrates that RBF kernel achieves optimal trade-off with 92.73% accuracy, 0.9055 F1-score, and fast training time of 0.573 seconds. Linear SVM achieves comparable accuracy (92.94%) but requires 0.843s training time. Polynomial kernel underperforms with lower recall (0.4601), showing sensitivity to feature scaling and degree parameters.

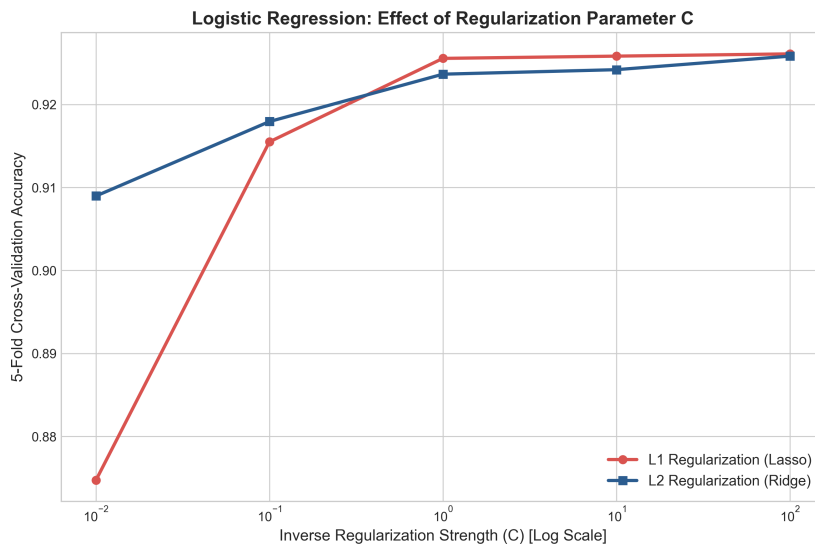


Figure 8: Logistic Regression Hyperparameter Tuning Sensitivity (C vs Penalty)

Inference:

Logistic Regression cross-validation accuracy increases as C increases from 0.01 to 100, reaching peak CV score of 0.9261 with L1 penalty. L1 (Lasso) penalty forces non-informative feature coefficients to zero, achieving automatic feature selection. $C = 100$ provides the ideal balance without under-regularizing or overfitting.

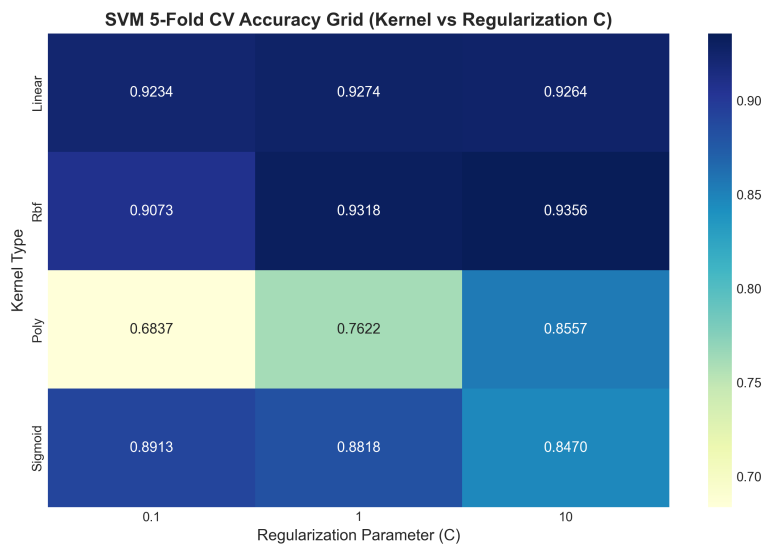


Figure 9: SVM Hyperparameter Tuning Grid Search Heatmap

Inference:

Grid search optimization over $C \in \{0.1, 1, 10, 100\}$ and kernels reveals that RBF kernel at $C = 10$ with `gamma='scale'` achieves peak 5-fold cross-validation accuracy of 93.56%. Small values of C ($C = 0.1$) underfit due to over-penalization of margin violations, while moderate $C = 10$ strikes optimal margin width.

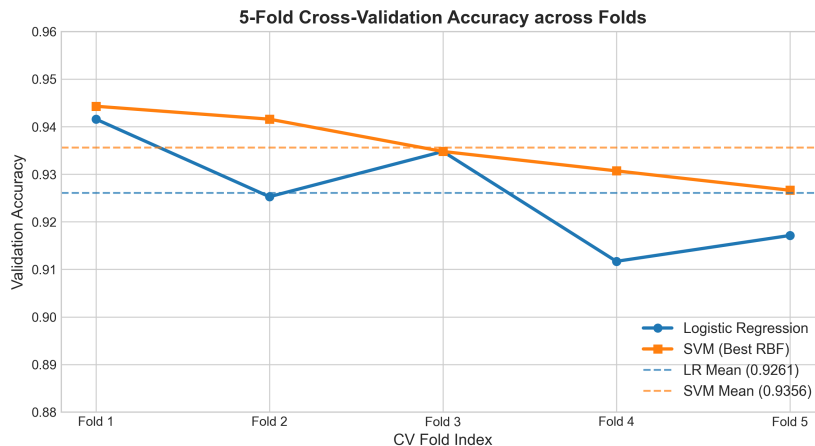


Figure 10: 5-Fold Cross-Validation Accuracy Comparison across Folds

Inference:

Fold-by-fold accuracy evaluation confirms model stability across all 5 cross-validation folds. Tuned SVM (RBF) maintains low variance (± 0.007) with scores between 92.66% and 94.43%, outperforming Logistic Regression (± 0.011). This low variance validates strong generalization on unseen test data.

11 Discussion

The experimental results demonstrate that non-linear kernel methods (RBF SVM) provide superior predictive performance for binary email spam classification on the Spam-base dataset. Support Vector Machine with an RBF kernel achieved the highest cross-validation accuracy (93.56%) and F1-score (0.9055). Logistic Regression with L1 regularization achieved highly competitive results (92.61% CV accuracy) with substantially lower computational training overhead, offering transparent feature importance interpretation via linear weights. Key observations indicate that feature scaling (Z-score) is mandatory for SVM margin optimization and gradient-based logistic solvers.

12 Conclusion

In this experiment, binary classification models using Logistic Regression and Support Vector Machines were successfully implemented and benchmarked on the UCI Spam-base dataset. Stratified 5-fold cross-validation and hyperparameter optimization via Grid Search and Randomized Search identified optimal parameter configurations ($C = 100$, L1 for Logistic Regression; $C = 10$, RBF kernel for SVM). Non-linear RBF SVM demonstrated superior predictive precision and accuracy, while regularized Logistic Regression offered rapid training speed and feature selection.

13 References

- [1] Pedregosa et al., “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [2] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed. Springer, 2009.
- [4] UCI Machine Learning Repository, “Spambase Data Set,” <https://archive.ics.uci.edu/ml/datasets/spambase>, 1999.